Surface Science 603 (2009) 653–658

Contents lists available at ScienceDirect

Surface Science

journal homepage: www.elsevier.com/locate/susc

Bulk and surface properties of spinel Co_3O_4 by density functional calculations

Xiang-Lan Xu [a], Zhan-Hong Chen [a,b], Yi Li [a], Wen-Kai Chen [a], Jun-Qian Li [a,*]

[a] Department of Chemistry, Fuzhou University, Qi Shan Campus, 2 Xue Yuan Road, University Town, Fuzhou 350108, PR China
[b] Department of Information Technology, Fujian Education College, Fuzhou 350001, PR China

ARTICLE INFO

Article history:
Received 7 October 2008
Accepted for publication 29 December 2008
Available online 21 January 2009

Keywords:
Spinel oxide
Co_3O_4
Bulk
(1 1 0) Surface
(1 1 1) Surface
DFT

ABSTRACT

DFT calculations are employed to bulk and surface properties of spinel oxide Co_3O_4. The bulk magnetic structure is calculated to be antiferromagnetic, with a Co^{2+} moment of 2.631 μ_B in the antiferromagnetic state. There are three predicted electron transitions $O(2p) \rightarrow Co^{2+}(t_{2g})$ of 2.2 eV, $O(2p) \rightarrow Co^{3+}(e_g)$ of 2.9 eV and $Co^{3+}(t_{2g}) \rightarrow Co^{2+}(t_{2g})$ of 3.3 eV, and the former two transitions are close to the corresponding experimental values 2.8 and 2.4 eV. The naturally occurring Co_3O_4 (1 1 0) and (1 1 1) surfaces were considered for surface calculations. For ideal Co_3O_4 (1 1 0) surfaces, the surface relaxations are not significant, while for ideal Co_3O_4 (1 1 1) surfaces the relaxation of Co^{2+} cations in the tetrahedral sites is drastic, which agrees with the experiment observation. The stability over different oxygen environments for possible ideal and defect surface terminations were explored.

© 2009 Published by Elsevier B.V.

1. Introduction

As an important magnetic p-type semiconductor, the spinel cobalt oxide Co_3O_4 is of special interest in a variety of technological applications and heterogeneous catalysts due to its surface redox reactivity properties. The Co_3O_4 as solid state sensor is reported to be sensitive to the isobutene [1], CH_4, H_2, NH_3, CO and NO_2 gases at low temperature [2]. Co_3O_4 shows high-catalytic activity for the oxidation of CO [3–7], N_2O catalytic decomposition [8]. Co_3O_4 nano-materials have been prepared and used as lithium batteries and gas sensors [9–11]. Several doped cobalt oxide spinels $M_xCo_{3-x}O_4$(M = Cu [12–14], Zn [15], Fe [16], Ni [17], Alkali metal [18–21]) have been proposed to enhance specific chemical or physical properties of the cobalt oxide spinel. Further understanding of the properties of Co_3O_4 bulk and its surface is crucial to potential development of the technological performance and efficiency of Co_3O_4-based materials applications.

Co_3O_4 is readily accessible and is the thermodynamically stable form of cobalt oxide under ambient room temperature and oxygen partial pressure. At room temperature, Co_3O_4 (Fig. 1) is a normal spine1 based on a cubic close-packing array of oxide ions where one-eighth of tetrahedral interstices are occupied by high-spin $Co^{2+}(d^7)$ ions, and one-half of octahedral interstices are occupied by low-spin $Co^{3+}(d^6)$ ions. As shown in (a) and (b) of Fig. 1, the tetrahedron $Co^{tet}O_4$ and octahedral $Co^{oct}O_6$ units are linked by sharing vertices, and the octahedral $Co^{oct}O_6$ units are linked by sharing edges. Co_3O_4 undergoes transition to an antiferromagnetic phase at Néel temperature of approximately 40 K. Roth reported the Co^{2+} moment in the antiferromagnetic state is 3.26 μ_B at 4.2 K [22].

Co_3O_4 single crystals generally form in a truncated octahedron morphology with naturally occurring (1 1 0) and (1 1 1) crystal faces [23]. Recently, the structures and chemical composition of the Co_3O_4 (1 1 0) and (1 1 1) surface have been studied by X-ray photoelectron spectroscopy (XPS), Auger electron spectroscopy (AES), low-energy electron diffraction (LEED), and scanning tunneling microscopy (STM) [24–27]. Like Fe_3O_4, there are two types of surface terminations for the (1 1 0) spinel orientation: type A which contains two Co^{2+} cations in the tetrahedral sites, two Co^{3+} cations in the octahedral sites and four O^{2-} anions and type B which has two Co^{3+} cations in the octahedral sites and four O^{2-} anions. The surface terminated by type A (or type B) is denoted as A (or B) surface. The LEED data have been shown to be consistent with type A termination, with Na, Ca, and K impurities segregated onto the surface [25]. Meyer et al. reported that the Co_3O_4 (1 1 1) oxide films deposited on Ir (1 0 0)-(1 × 1) in oxygen-rich conditions are terminated by a bulk-like layer of originally tetrahedrally coordinated Co ions [27].

The bulk and surface of spinel-type oxide studies have gradually intrigued researchers' attention in theory. A number of theoretical studies have been published on the bulk and surfaces of Fe_3O_4 [28–32], however, comparatively less attention has been given on the Co_3O_4. Broqvist et al. have studied CO oxidation mechanism on the Co_3O_4 (1 1 0) B surface using density functional theory (DFT) method modeled as an three-layer slab, and found that there is no significant displacement of the surface atoms along the (1 1 0) direction [33]. Piskorz et al. have performed a DFT molecular modeling of N_2O decomposition over Co_3O_4 (1 0 0) plane using a cluster approach [34].

* Corresponding author. Fax: +86 591 87892522.
E-mail address: jqli@fzu.edu.cn (J.-Q. Li).

0039-6028/$ - see front matter © 2009 Published by Elsevier B.V.
doi:10.1016/j.susc.2008.12.036

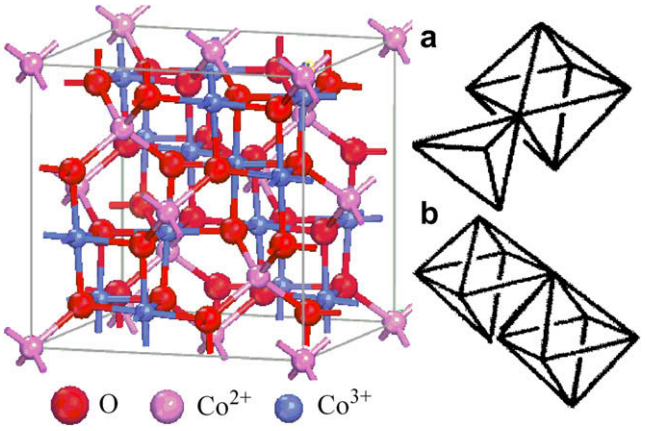


Fig. 1. The structure of the bulk Co_3O_4 and the link modes for tetrahedron $Co^{tet}O_4$ and octahedral $Co^{oct}O_6$ units.

In the present work, we investigated the bulk and surface properties of spinel Co_3O_4 by first-principles calculations based on spin-polarized DFT. The magnetic structure, band gaps, density of states and possible optical-transition bands of the bulk Co_3O_4 were studied. The structure, surface relaxation, and stability of possible ideal and defect Co_3O_4 (1 1 0) and (1 1 1) surfaces over different regions of oxygen chemical potential were also discussed.

2. Methods

2.1. Calculation details and models

All the calculations were performed with the DMol^3 [35] program package in Materials Studio of Accelrys Inc. In the computation, exchange–correlation effects were described by the generalized gradient approximation (GGA) developed by Perdew, Burke, and Ernzerhof (PBE) [36]. The molecular orbitals were expanded into a double-numerical basis with polarization functions (DNP). For O atoms, all electron basis sets were used. Effective core potentials were used for Co atoms, that is, the outer electrons ($3d^74s^2$) are treated as valence electrons and the remaining electrons are replaced by effective core potentials. A fermi smearing of 0.01 Hartree and a cutoff energy of 4.5 Å were used. The self-consistent field convergence criterion was set to be an energy change of 10^{-6} Hartree. The convergence criterion of optimal geometry based on the energy, force and displacement convergence, were 1×10^{-5} Hartree, 2×10^{-3} Hartree/Å and 5×10^{-3} Å, respectively. Throughout calculations the spin-polarized approach was used. The Brillouin zone integrations were performed using a $5\times5\times5$ Monkhorst–Pack grids for the bulk, $3\times4\times1$ for the (1 1 0) surface and $5\times5\times1$ for the (1 1 1) surface. We have tested the influence of the K points sampling with $6\times6\times6$ for the bulk, $4\times5\times1$ for (1 1 0) and $6\times6\times1$ for (1 1 1) surface. We found that the energies are not changed at the larger k-points. The Co_3O_4 (1 1 0) and (1 1 1) surfaces were simulated by using a symmetric periodic slab model. The slab was repeated periodically with a 12 Å vacuum region between the slabs. All the surfaces were fully optimized.

2.2. Surface energy

Following the approach developed by Reuter and Scheffler [37], for the surface system the surface free energy $\gamma(T,P)$ in an oxygen pressure p and temperature T is defined as

$$\gamma=\frac{1}{2A}\left[G^{slab}_{Co_3O_4}(T,P)-N_{Co}\mu_{Co}(T,P)-N_O\mu_O(T,P)\right], \quad (1)$$

where $G^{slab}_{Co_3O_4}(T,P)$ is the Gibbs free energy of the slab system; $\mu_{Co}(T,P)$ and $\mu_O(T,P)$ are the chemical potentials of Co and O atoms, respectively; A is the area of the surface unit cell; N_{Co} and N_O are the numbers of Co and O atoms in the slab. Because the surface system is modeled by a slab with two equivalent surfaces, the surface free energy is multiplied by a factor of 0.5. For the Co_3O_4 crystal, thermal equilibrium between surface and bulk requires that

$$3\mu_{Co}(T,P)+4\mu_O(T,P)=g^{bulk}_{Co_3O_4}(T,P), \quad (2)$$

where $g^{bulk}_{Co_3O_4}(T,P)$ is the Gibbs free energy of per Co_3O_4 formula unit. Inserting this constraint into Eq. (1) leads to

$$\gamma=\frac{1}{2A}\left[G^{slab}_{Co_3O_4}(T,P)-\frac{1}{3}N_{Co}g^{bulk}_{Co_3O_4}(T,P)+\left(\frac{4}{3}N_{Co}-N_O\right)\mu_O(T,P)\right], \quad (3)$$

which shows that the surface free energy depends only on the oxygen chemical potential.

The range of oxygen chemical potentials between our theoretical boundaries is

$$\frac{1}{4}\Delta G_f(0,0)\leqslant\mu_O(T,P)-\mu^{gas}_O\leqslant0 \quad (4)$$

The formation heat of the oxide, $\Delta G_f(T,P)$ is calculated by the Eq. (5):

$$\Delta G_f(T,P)=g^{bulk}_{Co_3O_4}(T,P)-3g^{bulk}_{Co}(T,P)-2g^{gas}_{O_2}(T,P), \quad (5)$$

where $g^{bulk}_{Co}(T,P)$ and $g^{gas}_{O_2}(T,P)$ is the Gibbs free energy of metallic cobalt solid and an O_2 molecule, and $\mu^{gas}_O=\frac{1}{2}g^{gas}_{O_2}(T,P)$.

In this paper, the Gibbs free energies were approximated by total energies at zero-temperature and zero-pressure environment from the DFT calculations, while the zero-point vibrations, vibrational entropy contributions, and enthalpy changes have been neglected for simplicity in the present calculation. This approximation has been used for the related inverse spinel system of the Fe_3O_4 (1 1 0) surface [32]. We compute $\Delta G_f(0,0)=-10.435$ eV per Co_3O_4 formula unit, which compares with the experimental Gibbs free energy at standard pressure and 298.15 K temperature, 8.022 eV [38]. In Eq. (4), the left boundary is denoted as oxygen-poor limit and the right boundary as oxygen-rich limit. In this paper, surface energies were calculated in the "oxygen-rich" and "oxygen-poor" boundaries.

3. Results and discussion

3.1. Calculations for the magnetic structure, band gaps, density of states and optical-transition bands of the bulk Co_3O_4

3.1.1. The magnetic structure and electronic band gaps

The bulk Co_3O_4 properties were initially performed to verify the reliability of the method in this paper. The lattice constant of the bulk Co_3O_4 is 8.084 Å [39]. Table 1 lists structural parameters, magnetic moment and electronic gap compared with the experimental data. The structural parameters are close to the experimental values. The spin population calculations show that the spin magnetic moment of a Co^{2+} ion is $2.631\mu_B$, and the Co^{3+} ion has a zero moment. The calculated moment of a Co^{2+} ion is smaller than the spin only value 3 μ_B and the experimental value 3.26 μ_B at 4.2 K for a Co^{2+} ion. We found that each Co^{2+} ion was surrounded by four nearest neighbor Co^{2+} ions of opposite spins. That is to say, the bulk Co_3O_4 shows antiferromagnetic properties, which is in a good agreement with the magnetic structure study by Roth [22].

The direct and indirect electronic band gaps at the two high-symmetry points Γ and X, which contribute to the valence and conduction band extreme, are listed in Table 1. Co_3O_4 has the cal-

Table 1
Comparisons between calculated and experimental data for bulk Co_3O_4.

Properties	Calc.	Exp.
$R_{Co}^{tet} - O^{2-}$ (Å)	1.933	1.935
$R_{Co}^{oct} - O^{2-}$ (Å)	1.921	1.920
Magnetic moment	2.631 μ_B/Co^{2+}	3.26 μ_B/Co^{2+}
Electronic gap (eV)	Γ–Γ = 2.06↑, 2.92↓ (1.67)[a]	0.83, 0.94, 1.7, 2.9 [41]
	X–X = 1.75↑, 2.23↓ (1.23)	1.10–2.06 [42]
	Γ–X = 1.92↑, 2.05↓ (1.51)	0.98, 1.10 [43]
		2.10 (Direct), 1.60 (Indirect) [44]

[a] The calculated values by DFT + U method are indicated in parentheses [40]. The symbols ↑ and ↓ denote the electrons of majority and minority states, respectively.

culated X–X direct gap of 1.75 eV, with an indirect Γ–X gap of 1.92 eV and a direct gap at Γ points of 2.06 eV. This is qualitatively in agreement with the calculated values by the DFT + U method [40] and thin films with variable thickness optical measurements [41–44].

3.1.2. Spin density of states and optical-transition bands of the bulk

The total and partial majority and minority spin density of states (DOS) of the bulk Co_3O_4 are shown in Fig. 2. For the Co^{2+} ions in the tetrahedral sites, the occupied majority spin d states are split between the e_g states at about −5 eV and the t_{2g} states at about −2 eV. The filled minority spin e_g states are situated at the top of the valence band, while the unoccupied minority t_{2g} states dominate the bottom of the conduction band at about +2 eV. These conduction band states of the Co^{2+} ions exhibit noticeable splitting due to the coupling with Co^{3+} at octahedral sites. The DOS contributed from the octahedral Co^{3+} sites is almost symmetric with respect to the majority and minority spin states, in agreement with crystal field and magnetic moment expectations in Fig. 3. The Co^{3+} 3d t_{2g} states are concentrated at the top of the valence band, about −1.3 eV. The e_g conduction band states are located at about +2.7 eV. The O 2p valence band spreads from −8 to 0 eV. The strong hybridization of the Co^{3+} 3d states with O 2p states results in the presence of O 2p states at the bottom of the conduction band, about +2.7 eV, while a weak hybridization for Co^{3+} 3d, Co^{2+} 3d and O 2p states are located at the top of the valence band, about −0.2 eV. The electronic density of states of the bulk Co_3O_4 calculated in this work are consistent with the previous DFT + U calculation using the same PBE functional implemented in VASP code [40].

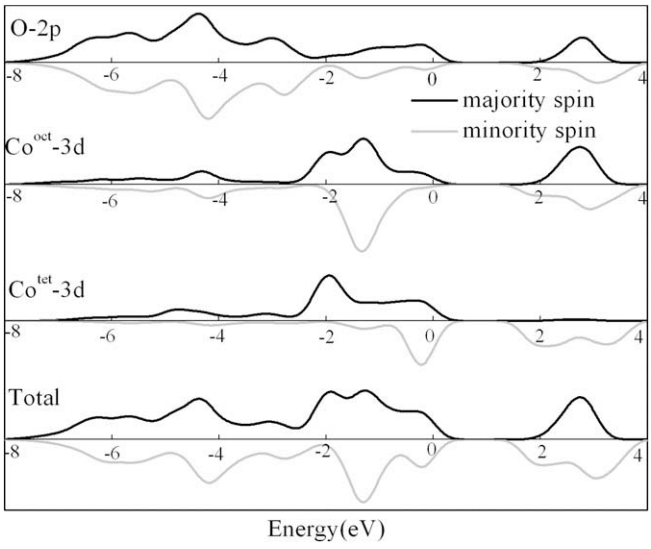


Fig. 2. The total and partial electronic density of states of the bulk Co_3O_4, plotted with reference to the valence band maximum at 0 eV.

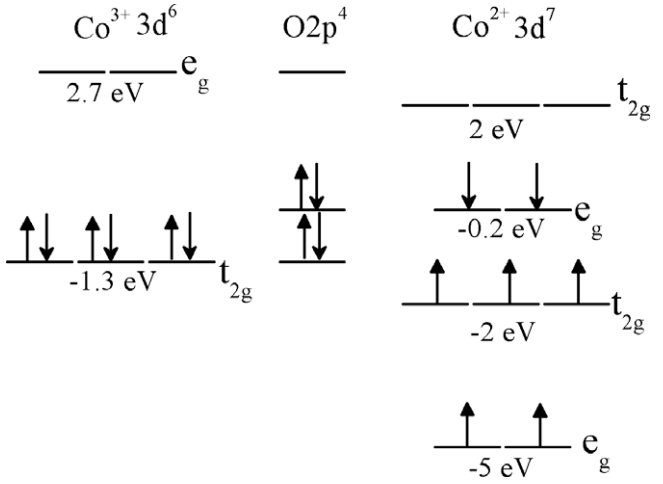


Fig. 3. Schematic crystal field splitting and occupations of the Co 3d electrons in tetrahedral and octahedral environments. The relative energy of Co(3d) and O(2p) derived from density of states calculations are indicated, with reference to the valence band maximum at 0 eV.

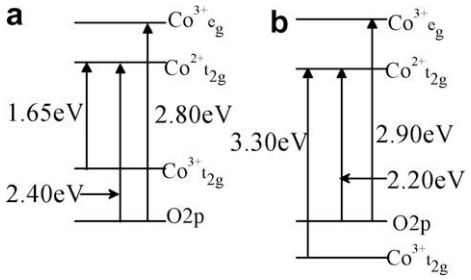


Fig. 4. The optic-transition structures (a) from the experiment [45] (b) in this work.

Athey et al. have reported that thin Co_3O_4 films exhibits strong optical absorption bands in the infrared–visible range [41]. Kim and Park have identified that there are three optical-transition structures located at about 1.65, 2.4, and 2.8 eV from the measured dielectric function of Co_3O_4 by spectroscopic ellipsometry (listed in Fig. 4a) [45]. Coupling with the analysis of the DOS in this work, the relative energy of Co(3d) and O(2p) are plotted in Fig. 3 with reference to the valence band maximum at 0 eV, and three possible electronic transition bands are predicted and plotted in Fig. 4b. The transition O(2p) → $Co^{2+}(t_{2g})$ for the 2.2 eV absorption and the transition O(2p) → $Co^{3+}(e_g)$ for the 2.9 eV absorption in this work are close to the corresponding values 2.4 and 2.8 eV in the experiment. While the transition $Co^{3+}(t_{2g}) \rightarrow Co^{2+}(t_{2g})$ for the 3.3 eV absorption is larger than 1.65 eV absorption in the experiment.

From these results of bulk calculations, we concluded that our theoretical approach is able to give a good description for the Co_3O_4 system.

3.2. Calculations for the ideal bulk terminated Co_3O_4 (110) surface

The composite, surface energy in the oxygen-rich conditions and the relaxation in the surface normal direction for Co_3O_4 (110)-A and -B surfaces as the function of layers were listed in Tables 2 and 3, respectively. The values Δz are the percentage changes of the fractional coordination in the surface normal direction after relaxation relative to bulk values. The coordination number of oxygen atoms is denoted as the superscript of the atom, for instance, O^{3f} is denoted as the three-fold oxygen atom.

Table 2
The composite, surface energy and relaxation in the surface normal direction for surface and subsurface atoms of the Co_3O_4 (1 1 0)-A surface as the function of layers.

Layers	Composite	C_o/C_{Co}	γ (J/m^2)	Δz/%					
				O^{3f}	Co^{oct}	Co^{tet}	O^{4f}	Co^{oct}	O^{4f}
3	$Co_{10}O_{12}$	1.20	2.561	0.6	−2.8	−4.5	3.1	0.0	3.9
5	$Co_{16}O_{20}$	1.25	2.564	1.1	−0.6	−2.0	0.3	1.6	0.7
7	$Co_{22}O_{28}$	1.27	2.568	1.1	−0.1	−0.9	0.6	1.5	1.0

Positive and negative values of Δz imply outward and inward displacements, respectively.

As the number of slab layers (N) increase, it can be found that the relaxation and the surface energy finally reach the convergence when N is up to 5 for A and B surfaces. For below discussed ideal and defect surface stability calculations (in Section 3.4), we chose the slab with five surface layers as the model. The Co_3O_4 (1 1 0)-A and -B surfaces with five surface layers are shown in Fig. 5.

Focusing on the surface relaxations of two surfaces in Tables 2 and 3, we note that the calculations predict no reconstruction. The relaxations in the surface normal direction for the surface and subsurface atoms are not significant, which is agreement with the theoretical calculations from Broqvist et al. [33]. After relaxation, the outmost surface atoms for A and B surfaces are always found to be O atoms (O^{3f} for the A surface and O^{2f} for the B surface). This might be due to the fact that the outmost atoms allow O anions, which attracts more valence electrons than the Co cations, to spread out electron clouds into the vacuum to reduce kinetic energy and accordingly to lower the system energy [31].

For the ideal Co_3O_4 (1 1 0)-A surface, surface O^{3f}, Co^{oct} and Co^{tet} atoms are in the same atomic monolayer. After relaxation, the surface A-termination splits into three layers with O atom being the outmost of surface, two Co^{oct} atoms situate at the second sub-layer and two Co^{tet} atoms at the third sub-layer. The equilibrium bond lengths of Co^{tet}–O^{3f} and Co^{oct}–O^{3f} are considerably reduced, from 1.935 to 1.863 Å and from 1.920 to 1.874 Å, respectively. The former is larger than the latter after relaxation, which is contrary to the case before relaxation. For the B surface, the equilibrium distance of Co^{oct}–O^{3f} bond (1.901 Å) is larger than Co^{oct}–O^{2f} bond (1.772 Å) after relaxation, compared with the same distance (1.920 Å) for them before relaxation. It is obvious that the bonding ability for the surface atoms is greatly strengthened due to the surface relaxation.

3.3. Calculations for the ideal bulk terminated Co_3O_4 (1 1 1) surface

Six inequivalent ideal bulk terminations can be obtained by cutting the spinel (1 1 1) stacking sequence, which are indicated in Fig. 6. They can expose Co^{tet}(Co^{tet1}, Co^{tet2}) or Co^{oct}(Co^{oct1}, Co^{oct2}), or close-packed oxygen double layers (O_1, O_2). There are two types of oxygen atoms for the oxygen double layer, and the ration of their atom number is 3:1 for the O_1 layer and 1:3 for the O_2 layer. A cleavage through the cobalt triple layer can generate two terminations consisting of one and two Co monolayers, respectively. This is clearly the most favorable cleavage since it only breaks five bonds against seven or nine for the cleavages between one oxygen layer and one Co layer. Based on the principle that the stable surfaces have the least dangling bonds, we chose the surface terminated with the Co^{oct2} and Co^{tet2} cations, and the surface with the Co^{tet1} cation as models (See Fig. 7). Thereafter, the two surfaces are denoted as A′ and B′ surfaces, respectively. The stacking sequences of the A′ and B′ surface with 1×1 unit cell are described by Co^{oct2}–Co^{tet1}–O_1–Co^{oct1}–O_2–Co^{tet2} and Co^{tet1}–O_1–Co^{oct1}–O_2–Co^{tet2}, respectively. The B′ surface model is consistent with the structure of the Co_3O_4 (1 1 1) oxide films deposited on Ir (1 0 0)-(1×1) in oxygen-rich conditions by quantitative low-energy electron dif-

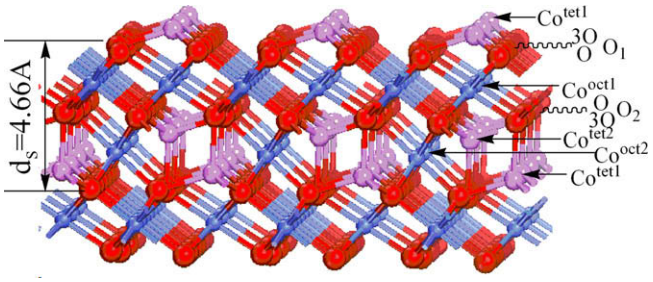


Fig. 6. The side view of the Co_3O_4 structure exposing a (1 1 1) surface plane. The vertical repeat length of layers d_s is 4.66 Å.

Table 3
The composite, surface energy and relaxation in the surface normal direction for surface and subsurface atoms of the Co_3O_4 (1 1 0)-B surface as the function of layers.

Layers	Composite	C_o/C_{Co}	γ (J/m^2)	Δz/%					
				O^{2f}	Co^{oct}	O^{3f}	Co^{tet}	Co^{oct}	O^{4f}
3	Co_8O_{12}	1.50	1.651	−3.1	−3.2	0.3	0	0	0
5	$Co_{14}O_{20}$	1.43	1.618	0.2	−1.4	0.2	3.6	0.6	0.1
7	$Co_{20}O_{28}$	1.40	1.614	0.5	−0.6	0.7	2.7	1.0	0.5

Positive and negative values of Δz imply an outward and inward displacement, respectively.

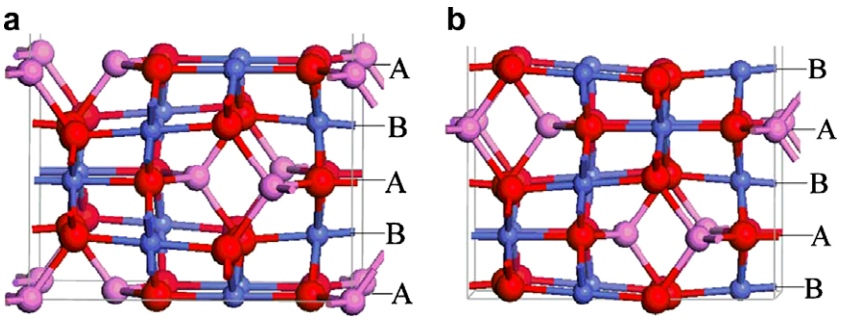


Fig. 5. Side view of the ideal Co_3O_4 (1 1 0) surfaces: (a) type A; (b) type B in 1×1 surface unit cell.

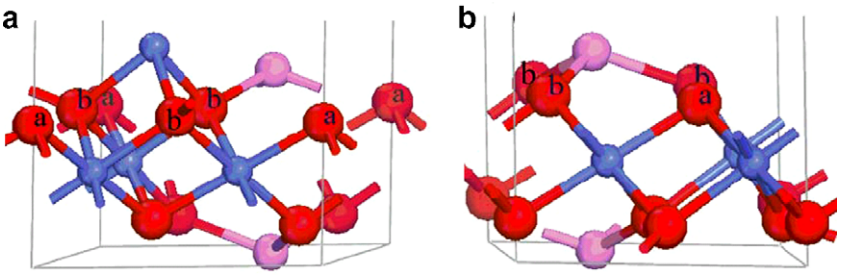


Fig. 7. Side view of the ideal Co_3O_4 (1 1 1) surfaces: (a) type A′; (b) type B′ in 1 × 1 surface unit cell.

Table 4
The composite, surface energy and relaxations in the surface normal direction and between interlayer for the top four atomic layers of the Co_3O_4 (1 1 1)-A′ and -B′ surfaces.

Surface	Composite	γ (J/m^2)	Interlayer/%		Δz/%			
			Co^{oct}–O_b	Co^{tet}–O_a	Co^{oct}	Co^{tet}	O_b	O_a
A′	Co_6O_8	1.779	−1.6	−70	0.55	−11.7	1.5	−0.7
Surface	Composite	γ (J/m^2)	Interlayer/%		Δz/%			
			Co^{tet}–O_b	O_a–Co^{oct}	Co^{tet}	O_b	O_a	Co^{oct}
B′	Co_5O_8	0.974	−32(50)[a]	−3.7(3.1)	−7.2	−1.5	−1.3	0

[a] The experimental values [27] are indicated in parentheses.

fraction (LEED) and scanning tunneling microscopy (STM) investigation [27].

Two inequivalent surface oxygen anions (O_a and O_b) yield after cleavage, which are indicated in Fig. 7. Single O_a is only coordinated to the Co^{oct} cation, and three O_b are coordinated to both Co^{oct} and Co^{tet} cations. The composite, surface energy in the oxygen-rich conditions and the relaxation for Co_3O_4 (1 1 1)-A′ and -B′ surfaces were shown in Table 4. We note that the surface energy of B′ surface is much smaller than A′ surface.

The relaxation calculations predict no reconstruction. For the two surfaces, the relaxations in the surface normal direction for Co^{oct}, O_b, O_a atoms are neglectable, and the relaxations for Co^{tet} atoms are significant. The Co^{tet} atoms is relaxed inward by 11.7 and −7.2 for Co_3O_4 (1 1 1)-A′ and -B′ surfaces, respectively. The interlayer relaxation results show that there is drastic relaxation of layer spacing between Co^{tet} and O_b, in the form of Co^{tet} relaxed inward towards the close-packed oxygen by 70 and 32 for Co_3O_4 (1 1 1)-A′ and -B′ surfaces, respectively. As a consequence, the equilibrium bond lengths between Co^{tet} and O_b atoms in the surface layer are greatly reduced, from 1.935 Å to 1.827 Å for the A′ surface and to 1.809 Å for the B′ surface. This is interpreted as accounting for the polarity compensation of the surface [27].

3.4. The stability calculations for Co_3O_4 (1 1 0) and (1 1 1) ideal and defect surfaces under oxygen-poor and -rich conditions

To compare the relative stabilities under different oxygen environment, the surface energies of two ideal and six defect models for the Co_3O_4 (1 1 0) surface, and two ideal and one defect models for the (1 1 1) surface as a function of oxygen chemical potential in the oxygen-rich and oxygen-poor conditions according to the Eq. (3) are presented in Fig. 8. For the defect (1 1 0) models, A–Co^{tet}–v, A–Co^{oct}–v and A–O^{3f}–v denote the A-terminated surface with one tetrahedral Co^{2+} vacancy, one octahedral Co^{3+} vacancy and one three-fold surface O atom vacancy, and B–Co^{oct}–v, B–O^{3f}–v and B–O^{2f}–v denote the B-terminated surface with one octahedral Co^{3+} vacancy, one three-fold and one two-fold surface O atom vacancy, respectively. For the defect (1 1 1) models, B′–O_a–v denotes B′-terminated surface with one O_a vacancy.

The surface energies of the A–Co^{tet}–v, A–Co^{oct}–v and A′ models (C_O/C_{Co} = 4/3) are independent of the oxygen chemical potential.

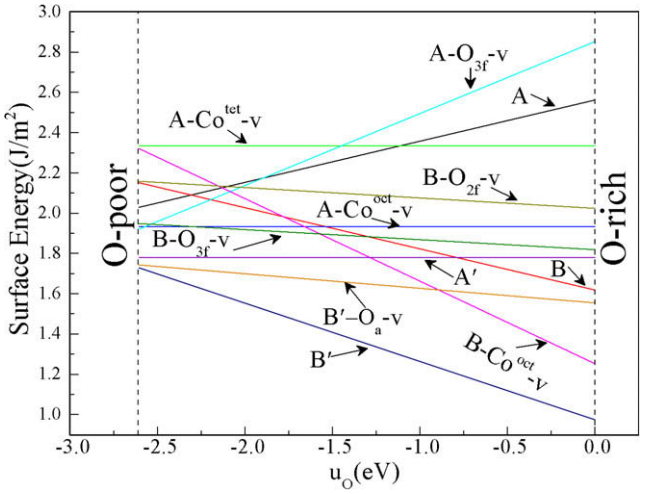


Fig. 8. Surface energies of the Co_3O_4 (1 1 0) and Co_3O_4 (1 1 1) surfaces as a function of the oxygen chemical potential, μ_O, using μ_O^{gas} as zero reference. The dotted vertical lines mark the oxygen-poor and oxygen-rich limits for μ_O following the same approach as that of Reuter and Scheffler.

With the increase of the oxygen chemical potential, the surface energies of B, B–Co^{oct}–v, B–O^{3f}–v, B–O^{2f}–v, B′, and B′–O_a–v models decrease, while that of A and A–O^{3f}–v models increase. In other words, B, B–Co^{oct}–v, B–O^{3f}–v, B–O^{2f}–v, B′, and B′–O_a–v models with the high-oxygen composition (C_O/C_{Co} > 4/3) concentration are more stable in an O-rich atmosphere. The A and A–O^{3f}–v models with the low-oxygen composition concentration (C_O/C_{Co} < 4/3) are more stable in an O-poor atmosphere. Among the supposed Co_3O_4 (1 1 0) surface models, the B–Co^{oct}–v and A–O^{3f}–v surface models are most probable to be formed in an O-rich atmosphere and in an O-poor atmosphere, respectively. For Co_3O_4 (1 1 1) surface models, the B′ surface is more stable in both O-rich and O-poor atmospheres.

4. Conclusions

In the present article, we have carried out a detailed spin-polarized density functional theory (DFT) investigation of the properties

of the bulk and surface of the spinel oxide Co_3O_4. The magnetic structure, band gaps, density of states and possible optical-transition bands of the bulk, and structure, relaxation and stability of Co_3O_4 (1 1 0) and (1 1 1) surface were studied.

The antiferromagnetic properties with each Co^{2+} ion in the A site surrounded by four nearest neighbor Co^{2+} ions of opposite spins, which is in line with the Roth's description of magnetic structure. The calculated X–X direct gap, Γ–X indirect gap and Γ–Γ direct gap is qualitatively in agreement with the calculated values by the DFT + U method [40] and thin films with variable thickness optical measurements [41–44]. Especially, the X–X direct gap 2.06 eV is consistent with the experiment values 2.06 eV [42] or 2.10 eV [44].

The filled minority spin e_g states of Co^{2+} ions are present at the top of the valence band, while its unoccupied minority t_{2g} states dominate the bottom of the conduction band at about +2 eV. The Co^{3+} 3d t_{2g} states are concentrated at the top of the valence band, about −1.3 eV. The e_g conduction band states are located at about +2.7 eV. The strong hybridization of the Co^{3+} 3d states with O 2p states results in the presence of O 2p states at the bottom of the conduction band, about +2.7 eV. There are three electron transitions $O(2p) \rightarrow Co^{2+}(t_{2g})$, $O(2p) \rightarrow Co^{3+}(e_g)$ and $Co^{3+}(t_{2g}) \rightarrow Co^{2+}(t_{2g})$ are predicted.

For ideal Co_3O_4 (1 1 0) surfaces, the results predict that the surface relaxations are not significant. After relaxation, the outmost surface atoms for A and B surfaces are always found to be O atoms, which is same as the case of the Fe_3O_4 (1 1 0) surfaces [32]. Surface energy calculations demonstrate that the A surface is easier to form in an O-poor atmosphere and the B surface in an O-rich atmosphere. For ideal Co_3O_4 (1 1 1) surfaces, we found that the B′ surface is more stable than the A′ surface. For the two (1 1 1) surfaces, the relaxation in the surface normal direction for Co^{tet} atoms is obvious, which agree with the experiment investigation [27]. Among the supposed two ideal and six defect Co_3O_4 (1 1 0) surface models, the B surface with octahedral cobalt cations vacancy is easiest to form in an O-rich atmosphere and the A-terminated surface with the three-fold surface oxygen anion vacancy in an O-rich atmosphere. Among the Co_3O_4 (1 1 1) surfaces, the B′ surface terminated with the Co^{tet} cation is most stable.

Acknowledgments

We kindly acknowledge financial support from the Natural Science Foundation of China (20673019, 20303002), the Key Project of the Fujian Province (2005HZ01-2-6), the Foundation of State Key Laboratory Breeding Base of Photocatalysis of Fuzhou University (No. K-081001), and the Natural Education Department Foundation of PhD Unit (20050386003).

References

[1] S.D. Choi, B.K. Min, Sens. Actuators B 77 (2001) 330.
[2] J. WÖllensteina, M. Burgmairb, G. Pleschera, T. Sulima, J. Hildenbranda, H. BÖttnera, I. Eiseleb, Sens. Actuators B 93 (2003) 442.
[3] J. Jansson, M. Skoglundh, E. Fridell, P. Thormählen, Top. Catal. 16–17 (2001) 385.
[4] H.-K. Lin, C.-B. Wang, H.-C. Chiu, S.-H. Chien, Catal. Lett. 86 (2003) 63.
[5] H.-K. Lin, H.-C. Chiu, H.-C. Tsai, S.-H. Chien, C.-B. Wang, Catal. Lett. 88 (2003) 169.
[6] R. Sundararajan, V. Srinivasan, Appl. Catal. A 141 (1996) 45.
[7] Z. Zhang, H. Geng, L. Zheng, B. Dua, J. Alloys Compd. 392 (2005) 317.
[8] N. Russo, D. Fino, G. Saracco, V. Specchia, Catal. Today 119 (2007) 228.
[9] Y. Li, B. Tan, Y. Wu, Nano Lett. 8 (2008) 265.
[10] Y. Li, B. Tan, Y. Wu, J. Am. Chem. Soc. 128 (2006) 14258.
[11] W.-Y. Li, L.-N. Xu, J. Chen, Adv. Funct. Mater. 15 (2005) 851.
[12] W.M. Shaheena, A.A. Ali, Mater. Res. Bull. 36 (2001) 1703.
[13] A. LaRosa-Toro, R. Berenguer, C. Quijada, F. Montilla, E. Morallon, J.L. Vazquez, J. Phys. Chem. B 110 (2006) 24021.
[14] P. Stefanov, I. Avramova, D. Stoichev, N. Radic, B. Grbic, T. Marinova, Appl. Surf. Sci. 245 (2005) 65.
[15] L. Yan, T. Ren, X. Wang, Q. Gao, D. Ji, J. Suo, Catal. Commun. 4 (2003) 505.
[16] K.J. Kim, H.K. Kim, Y.R. Park, G.Y. Ahn, C.S. Kim, J.Y. Park, J. Magn. Magn. Mater. 300 (2006) 300.
[17] L. Yan, T. Ren, X. Wang, D. Ji, J. Suo, Appl. Catal. B 45 (2003) 85.
[18] M. Haneda, Y. Kintaichi, N. Bion, H. Hamada, Appl. Catal. B 46 (2003) 473.
[19] M. Haneda, Y. Kintaichi, H. Hamada, Appl. Catal. B 55 (2005) 169.
[20] C. Ohnishi, K. Asano, S. Iwamoto, K. Chikama, M. Inoue, Catal. Today 120 (2007) 145.
[21] K. Asano, C. Ohnishi, S. Iwamoto, Y. Shioya, M. Inoue, Appl. Catal. B 78 (2008) 242.
[22] W.L. Roth, J. Phys. Chem. Solids 25 (1964) l.
[23] J.L. Hutchison, N.A. Briscoe, Ultramicroscopy 18 (1985) 435.
[24] S.C. Petitto, M.A. Langella, J. Vac. Sci. Technol. A 22 (2004) 1690.
[25] S.C. Petitto, M.A. Langell, Surf. Sci. 599 (2005) 27.
[26] S.C. Petitto, E.M. Marsh, G.A. Carson, M.A. Langell, J. Mol. Catal. A: Chem. 281 (2008) 49.
[27] W. Meyer, K. Biedermann, M. Gubo, L. Hammer, K. Heinz, J. Phys.: Condens. Mat. 20 (2008) 265011.
[28] M.E. Grillo, M.W. Finnis, W. Ranke, Phys. Rev. B 77 (2008) 075407.
[29] L. Zhu, K.L. Yao, Z.L. Liu, Phys. Rev. B 74 (2006) 035409.
[30] D.M. Huang, D.B. Cao, Y.W. Li, H.J. Jiao, J. Phys. Chem. B 110 (2006) 13920.
[31] C. Cheng, Phys. Rev. B 71 (2005) 052401.
[32] Y.L. Li, K.L. Yao, Z.L. Liu, Surf. Sci. 601 (2007) 876.
[33] P. Broqvist, I. Panas, H. Persson, J. Catal. 210 (2002) 198.
[34] W. Piskorz, F. Zasada, P. Stelmachowski, A. Kotarba, Z. Sojka, Catal. Today 137 (2008) 418.
[35] B. Delley, J. Chem. Phys. 113 (2000) 7756.
[36] J.P. Perdew, K. Burke, M. Ernzerhof, Phys. Rev. Lett. 77 (1996) 3865.
[37] K. Reuter, M. Scheffler, Phys. Rev. B 65 (2001) 035406.
[38] D.R. Lide, CRC Handbook of Chemistry and Physics, CRC Press, Boca Raton, 2004.
[39] X. Liu, C.T. Prewitt, Phys. Chem. Miner. 17 (1990) 168.
[40] A. Walsh, S.-H. Wei, Y. Yan, M.M. Al-Jassim, J.A. Turner, Phys. Rev. B 76 (2007) 165119.
[41] P.R. Athey, F.K. Urbanill, M.F. Tabet, W.A. McGahan, J. Vac. Sci. Technol. A 14 (1996) 685.
[42] P.S. Patil, L.D. Kadam, C.D. Lokhande, Thin Solid Films 272 (1996) 29.
[43] L.D. Kadam, P.S. Patil, Mater. Chem. Phys. 68 (2001) 225.
[44] V.R. Shinde, S.B. Mahadik, T.P. Gujar, C.D. Lokhande, Appl. Surf. Sci. 252 (2006) 7487.
[45] K.J. Kim, Y.R. Park, Solid State Commun. 127 (2003) 25.